Assignment 12

Title: Industry Capstone: - End-to-End NLP Application

Project Title: AQI Health Query Classifier

Abstract

Air pollution is a significant environmental and public-health concern. The Air Quality Index (AQI) communicates pollution levels and their potential health effects, but conventional AQI dashboards mainly display numerical values and general health advisories. Users may still need help identifying information relevant to their specific concerns.

The AQI Health Query Classifier is an NLP-based machine-learning application that classifies natural-language queries related to air quality, pollution exposure, and human health into nine predefined intent categories. The system preprocesses text using normalization, tokenization, stop-word removal, and lemmatization. It then extracts features using TF-IDF and applies trained classification models to predict the query category. The application also provides confidence scores, model comparisons, educational advisories, analytical visualizations, and persistent query history.

The system was evaluated using accuracy, precision, recall, and F1-score. The Calibrated Linear SVM achieved 92.55% accuracy on the held-out test set, making it the best-performing model among the evaluated approaches. The application is intended for educational and health-awareness purposes and does not replace professional medical advice.

Keywords: Air Quality Index, Natural Language Processing, Machine Learning, TF-IDF, Text Classification, Health Queries.

1. Introduction

Air pollution can affect human health, particularly respiratory health and the well-being of vulnerable groups. AQI dashboards help users understand air-quality conditions through numerical values, pollutant measurements, and severity indicators. However, these dashboards may not directly address individual questions such as whether outdoor exercise is advisable, how pollution can affect breathing, or what precautions can reduce exposure.

The AQI Health Query Classifier addresses this problem by allowing users to enter questions in ordinary language. Instead of relying entirely on fixed menus or exact keyword matching, the system uses NLP and machine learning to identify the likely intent of a query and display relevant general health-awareness information.

1.1 Problem Statement

Conventional AQI information systems generally provide numerical readings and predefined advisories. They do not necessarily classify free-form questions according to the user's specific concern. The proposed system introduces an automated query-classification mechanism to connect natural-language questions with appropriate information categories.

1.2 Objectives

- To develop an NLP pipeline for processing AQI and health-related queries.
- To classify queries into nine predefined intent categories.
- To train and compare multiple machine-learning models.
- To display prediction confidence and relevant educational advisories.
- To evaluate model performance using standard classification metrics.
- To provide an interactive interface for analysis, simulation, and query-history management.

2. Proposed Methodology

The system uses a labelled dataset containing **1,272 queries across nine categories**: AQI Information, Health Effects, Symptoms, Precautions, Masks and Protection, Outdoor Activities, Vulnerable Groups, Respiratory Health, and Pollution Exposure.

The processing pipeline begins when a user submits a natural-language query. The text is normalized, tokenized using NLTK, stripped of stop words, and lemmatized using WordNet resources. Domain-specific tokens such as PM2.5 and N95 are preserved in normalized form to retain their meaning.

The processed text is converted into numerical features using TF-IDF with unigram and bigram features. The project evaluates four classifiers: Logistic Regression, Calibrated Linear SVM, Random Forest, and a Deep Self-Attention neural network implemented using PyTorch.

The selected model predicts the intent category and provides confidence information. Based on the predicted category, the application retrieves relevant educational guidance and displays an appropriate health disclaimer. Model performance is evaluated using a held-out test set and metrics such as accuracy, weighted precision, weighted recall, F1-score, and a confusion matrix.

3. System Implementation

The application is developed in Python using Streamlit for the user interface, Scikit-learn for classical machine-learning models, PyTorch for the deep-learning model, NLTK for text preprocessing, and SQLite for persistent query-history storage.

The main application components are:

1. **Dashboard:** Displays key statistics, category distribution, and recent query activity.
2. **Query Analyzer:** Accepts user questions and displays the predicted category, confidence score, top-three predictions, preprocessing details, feature explanations, and educational advisory.
3. **AQI Simulator:** Allows users to explore illustrative AQI and pollutant scenarios through interactive controls and visualizations.
4. **Analytics Dashboard:** Presents evaluation metrics, confusion matrices, and category-wise performance charts.
5. **Model Comparison:** Compares the evaluated classifiers using performance metrics and other model characteristics.
6. **Query History:** Stores query records in SQLite and supports searching, filtering, deletion, and CSV export.
7. **Dataset Explorer:** Displays raw and preprocessed dataset samples, category distributions, and related visualizations.

The system also provides input validation and an explicit disclaimer that its output is for general awareness rather than medical diagnosis.

4. Results and Discussion

The evaluated models produced the following results on the held-out test set:

Model	Accuracy	Weighted F1-score
Calibrated Linear SVM	92.55%	92.60%
TF-IDF + Logistic Regression	92.16%	92.17%
Deep Self-Attention	86.67%	86.59%
TF-IDF + Random Forest	83.14%	83.11%

The Calibrated Linear SVM achieved the highest accuracy and weighted F1-score among the four evaluated models. Logistic Regression also performed competitively, while the Deep Self-Attention and Random Forest models obtained lower scores on the same evaluation setup.

These results demonstrate that classical TF-IDF-based classification can be effective for the project's short-text intent-classification task. However, performance may vary for ambiguous questions, unfamiliar vocabulary, or queries that fall outside the nine defined categories. The results should therefore be interpreted in the context of the available dataset and evaluation procedure.

5. Conclusion and Future Scope

The AQI Health Query Classifier demonstrates how NLP and machine learning can be applied to classify natural-language questions about air quality and health. By combining text preprocessing, feature extraction, multiple classification models, confidence information, and educational advisory retrieval, the system provides a structured alternative to relying solely on static AQI information displays.

The Calibrated Linear SVM achieved the best measured performance, with 92.55% accuracy. The interactive dashboard, model analytics, AQI simulator, dataset explorer, and SQLite-based query history provide additional functionality for demonstration and evaluation.

Future enhancements may include multilingual query support, integration with verified real-time AQI data sources, improved handling of ambiguous or multi-intent queries, and evaluation using larger and more diverse real-world datasets. The application remains a health-awareness tool and is not intended to provide medical diagnosis or replace professional healthcare guidance.